



The HapBlockR Breeder's Guide

Choosing Parents and Crosses with Haplotype-Aware Tools — A Plain-Language Companion

By Félicien Akohoue, PhD

Quantitative Genetics

1st Edition

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About this guide

HapBlockR gives a breeding programme eight tools for turning a ranked list of candidates into an actual crossing decision. Each tool answers a different question — "**who are my best parents**," "**which parents together cover what I actually want covered**," "**which specific crosses are worth making**," "**how much should each parent contribute, and who mates whom, with inbreeding under control**" — and none of them is a strict upgrade on the others. Picking the right one for the decision in front of you matters more than knowing every statistical detail behind it.

This guide is written for breeders, programme managers, and reviewers who will use or approve a crossing decision but do not necessarily run R themselves. For every tool it explains: **what it is**, **what question it answers**, **a breeding-programme analogy**, **when to reach for it**, **when to be cautious or skip it**, and **how it behaves in practice** — with no R code anywhere in this document. If you do work in R, or want the full statistical detail, equations, and worked examples, see the companion vignette *From Local GEBV to a Crossing Decision* (run `vignette("HapBlockR-breeding-decisions", package = "HapBlockR")`), or find it on the package's documentation site).

A note on this edition. This guide has been substantially expanded from earlier versions. It now covers eight tools instead of seven — the new addition is family- and genetic-cluster-quota selection (`select_parents_by_family()`) — and the genetic-algorithm parent selection section has been expanded to cover its two optional "dials" for merit and relatedness, including the easier 0-90 and 0-100 scales that were added since the previous edition.

Part 1 — The big picture

1.1. The pipeline this toolkit sits inside

Every tool in this guide sits downstream of one starting point: you already have a breeding value or selection index for every candidate parent, from whatever prediction method your programme trusts (a mixed model, GBLUP, ASReml-R, or any other approach). HapBlockR does not replace that step. What it adds is everything between "here is a ranked list of candidates" and "here is next season's crossing block":

1. **Where** in the genome that breeding value is coming from — not just one genome-wide number, but a breakdown by haplotype block, so you can see which specific regions are driving performance.
2. **Which set** of parents, taken together, covers the most favourable haplotype blocks — and how that differs from simply taking the highest-ranked individuals one at a time.
3. **Whether that set is balanced** across your programme's families, or whether one family or genetic group has quietly taken over the crossing block.
4. **Which specific crosses**, among your shortlisted parents, are worth making — not just which parents to keep.

5. **How much each parent should contribute**, and which matings should happen, with population-wide inbreeding held under an explicit, negotiated cap — not just checked as an afterthought.

The eight tools in this guide answer pieces of that chain. However, none of them replaces your breeding-value prediction step, and none of them removes the need for a breeder's judgement — they narrow the search, make trade-offs visible, and give you evidence for a decision instead of an unexplained ranked list.

1.2. A map of the eight tools

Stage of the decision	Tool	What it decides
Simplest shortlist	Truncation Selection	Rank by one score, take the top N
Block-aware shortlist	GA Parent Selection	Which <i>set</i> of parents jointly covers your target haplotype blocks
Group-balanced shortlist	Family- or Genetic-Cluster-Quota Selection	Best few groups, then best few lines within each
Cross ranking	Usefulness Criterion	Which specific pairings are worth making
Full mating plan	Optimal Contribution Selection	How much each parent contributes, and who mates whom, under an inbreeding cap
Seeing the trade-off	Pareto Frontier Sweep	The actual gain-vs-relatedness curve, not one guessed number
Sanity check	Exact Validation	How close a heuristic mating plan is to the true best achievable one
Diversity-first selection	Core Collection	A representative, diverse subset when diversity itself — not merit — is the goal

GA = *genetic algorithm*.

Part 2 walks through each of these in turn. Part 3 gives a consolidated decision guide and a glossary of the recurring technical terms.

1.3. Two layers, not one straight line

It's easy to read the table above as a single pipeline — Truncation Selection, then GA Parent Selection, then Family-Quota Selection, then Usefulness Criterion, then Optimal Contribution Selection — and that is not quite right. There are two genuinely separate layers, and understanding the difference will save you from misreading how these tools are meant to be combined:

- **The shortlist layer.** Truncation Selection, GA Parent Selection, and Family- or Genetic-Cluster-Quota Selection are *alternatives to each other*, not steps in a sequence. Each answers "**who are**

my candidate parents" using different logic. The normal way to use them is to run more than one and compare — not to feed one's output into another as if it were a refinement step.

- **The downstream layer.** Usefulness Criterion (cross ranking) and Optimal Contribution Selection (the full mating plan) both sit one stage further on, and neither one builds its own candidate list. Both are built to take *whatever* shortlist you settled on — from any of the three tools above, or a list you curated by hand — and work from that. Neither cares which upstream tool produced the list it was handed.

A related point worth being explicit about: Usefulness Criterion and Optimal Contribution Selection are not a sequence either. They are siblings at the same downstream layer, answering different questions from the same shortlist — cross ranking scores individual candidate pairs with no notion of an overall plan, while Optimal Contribution Selection solves a different, population-level problem (contributions and matings under an explicit relatedness cap) directly from your shortlist's merit values and a relationship matrix. You can use a cross ranking to get a feel for which pairings look promising before running a full mating-plan tool, but the mating-plan tool does not require the cross ranking as an input — it does not read or depend on it in any way.

Pick a shortlist strategy (or run more than one and compare), then hand whichever shortlist you're using to whichever downstream question you're asking — cross ranking, a full mating plan, or both, independently.

One qualification worth knowing before you rely on this. "Downstream, takes whatever shortlist you hand it" is accurate about *where candidates come from* — the mating-plan tool can never pull in a parent that wasn't in the list you gave it. It is not the same as "the mating plan will use every candidate in that list." None of the three underlying solvers is obliged to give every supplied candidate a role in the final plan, and one of them — the AlphaMate engine — can narrow the list further as a deliberate, built-in part of solving the mating problem itself, if you ask it to cap the number of parents used. In that specific situation, the "mating plan" step is also doing some of the "who are my parents" work, on top of whatever you already shortlisted. Section 2.5 explains this in full.

1.4. A few ideas worth understanding before the rest of the guide

A handful of concepts recur across almost every tool in this guide. Getting comfortable with them here makes the rest of the guide easier to follow.

Local GEBV (Genomic Estimated Breeding Value) / haplotype blocks. A single genome-wide breeding value tells you *how good* a candidate is, not *where* that value is coming from. HapBlockR's prediction step re-expresses the same value as a sum of contributions from individual haplotype blocks — short, co-inherited runs of the genome. This matters because two candidates can have the same overall breeding value while carrying completely different favourable regions; a tool that only sees the whole-genome number cannot tell the difference, but a tool that sees the block-level breakdown can.

A relationship matrix (G). Several tools in this guide need a way to measure how related any two candidates are — full sibs are more related than half sibs, which are more related than unrelated lines from different families. This is captured in a relationship matrix (sometimes called a genomic relationship matrix, or GRM), built from marker data or pedigree records. A higher value between two individuals means they are more closely related; the diagonal reflects an individual's own inbreeding. Several dials in this guide (target degree, coancestry weight, relatedness ceiling) are built directly from this matrix.

Merit vs. diversity — the recurring trade-off. Almost every advanced tool in this guide manages some version of the same tension: pushing hard for the highest possible merit tends to concentrate selection on a small number of related, top-performing lines, which increases inbreeding risk down the line; pushing hard for diversity spreads the selection out but can leave real gain on the table. None of the tools in this guide resolve that tension for you — they make it visible and adjustable, on a shared 0-90 scale wherever possible (0 = prioritise merit/gain, 90 = prioritise diversity/minimise relatedness), so that a decision you make with one tool carries the same meaning when you make it with another.

Shrinkage / small-sample correction. A newer idea in this guide (used by family-quota selection) borrows a standard quantitative-genetics correction: a small family's raw average is a noisier estimate of its true quality than a large family's raw average is, even if the two numbers come out identical on paper. Shrinkage pulls a small group's estimate part-way toward the overall average, in proportion to how little that group's own data can be trusted — the smaller the group, the harder the pull. This avoids letting a single lucky (or unlucky) small family jump to the top (or bottom) of a ranking purely because it has too few members to trust its own average at face value.

Part 2 — The eight tools

1.5. Truncation Selection — the baseline every smarter method has to beat

What it is. The simplest possible parent-selection rule: rank every candidate by one number — a breeding value, a selection index, anything you already trust — and take the top N. Nothing more.

What question it answers. "Who scores the highest, one at a time?"

Breeding-programme analogy. This is the equivalent of ranking every candidate by GPA and admitting the top of the class, with no regard for which subjects each candidate is strong in or how many students from the same high school get in.

When to reach for it.

- You have one already-validated selection index and no open question about whether your shortlist is missing complementary genetic material.
- Your programme is early-stage or does not yet have the infrastructure for a full haplotype-block workflow.

- You are under real time pressure — a release deadline, a seed increase — where a fast, transparent, easy-to-explain shortlist matters more than incremental optimization.
- Always, as a baseline. Every smarter tool in this guide should be run alongside truncation selection at least once, so you have direct evidence that the extra machinery changed the outcome for the better, not just added complexity.

When to be cautious or look further instead.

- If your elite candidates cluster tightly on a handful of pedigrees, a top-N-by-value list is very likely to also be a top-N-by-relatedness list — the family/diversity check described in Section 2.3 exists to catch this after the fact, but if you already expect it, consider a group-balanced or block-aware tool instead.
- If you have already identified specific favourable haplotype regions worth deliberately combining, truncation selection has no mechanism to act on that information at all — it treats every point of merit as interchangeable, wherever in the genome it comes from.

How it works in practice. One input list (your trusted score), one number (how many parents you want), and — optionally — a floor that excludes anyone below a minimum bar before ranking begins. No relationship matrix, no block data, no tuning required. The output is a plain list of chosen individuals, ready to use immediately or as the comparison arm for every other tool in this guide.

1.6. GA Parent Selection — optimizing for joint block coverage

What it is. A search — powered by a genetic algorithm, the same optimization approach used across many engineering and logistics problems — for the *set* of parents that, taken together, covers the most favourable value across a chosen list of important haplotype blocks. This is a fundamentally different question from truncation selection's "who scores highest individually."

What question it answers. "Which combination of parents, considered as a group, gives me the best coverage of the specific genetic regions I've decided matter most?"

Breeding-programme analogy. Instead of admitting the highest GPA students one at a time, this is closer to assembling a team where every required skill is covered by at least one member — even if that means passing over an individually stronger candidate whose skills entirely duplicate someone already on the team.

Why this catches something truncation selection cannot. The top individuals by whole-genome value can, purely by chance, be concentrated on the *same* favourable haplotype blocks — leaving other genetically important regions completely unrepresented in the shortlist. GA parent selection searches specifically for a group that avoids this blind spot.

Crossing-scheme awareness. The tool understands that "covering" a block means different things under different crossing schemes: sometimes two distinct parents are needed to jointly deliver a block's value (a standard biparental cross); sometimes one parent alone is enough (selfing, or a doubled-haploid-style single-parent gamete donation); sometimes any single parent carrying the block counts, with no specific pairing required (a population-value framing). You tell the tool which of these matches your actual programme, and the search respects it.

When to reach for it.

- Once you have a specific list of target haplotype blocks worth stacking simultaneously — this is the tool for "which set of individuals, taken together, covers what I actually want covered."
- Your elite candidates carry overlapping favourable regions, so a plain ranked list would double up on the same blocks and leave others uncovered.
- You are assembling a multi-parent founder population where joint complementarity matters more than any single individual's rank.
- You need a specific crossing-scheme assumption enforced *during* the search, not checked afterward.

When to skip it, for now.

- You need the block-level breakdown from the earlier prediction step first — there is no version of this tool that works from a single whole-genome score alone (that's Section 2.1).
- For very large candidate pools without pre-filtering to a manageable subset, search time becomes a real practical cost worth budgeting for.
- If a plain truncation-selection list already looks complementary (this tool's own overlap check tells you this directly), the extra workflow step may not change the outcome enough to justify running it every cycle.

Two optional dials, for two common concerns. Two further refinements address gaps in the plain block-coverage search:

- **A merit dial**, for when block-coverage estimates are noisy. Per-block estimates are themselves statistical predictions, not ground truth. Left unchecked, a coverage-only search can lean on a candidate whose apparent block coverage is partly a noisy artefact, if that candidate clears a minimum bar at all. Turning on the merit dial makes whole-genome merit compete directly with block coverage throughout the search, not just as an entry gate. Rather than guessing a raw weighting number, a simpler 0-100 "how much do I care about merit vs. coverage" percentage is available, auto-calibrated from your own data — 0 leaves the search exactly as coverage-only, 100 gives merit as much realistic influence on the outcome as coverage itself has.
- **A relatedness dial**, for when the chosen founder set risks being too closely related. By default, this tool has no awareness of relatedness at all — two chosen parents could be full sibs. Supplying a relationship matrix unlocks a relatedness penalty inside the search itself. As with the merit dial, rather than guessing a raw penalty number, a 0-90 scale is available (the same scale used consistently across this whole toolkit): 0 prioritises coverage/merit and accepts more relatedness; 90 prioritises minimizing relatedness. This dial converts your stated preference into an actual relatedness ceiling — a real bound, not an abstract multiplier — built from two genuinely constructed reference groups for your own candidate pool: a high-coverage reference at one end, and a proven diversity-maximizing reference at the other.

Checking the search is trustworthy. A single run of a stochastic search like this says nothing on its own about whether the answer is a robust optimum or one of several near-equally-good solutions the search happened to land on. By default, the search is repeated several times from different random starting points, and reports what fraction of those repeats agreed on each selected individual — candidates chosen every single time are robustly supported; candidates chosen only once are borderline and worth a second look before committing to them.

How it works in practice. You need the block-level breakdown from the prediction step, the number of founders you want, your crossing-scheme assumption, and (recommended) a per-block weighting reflecting how important each block is. Always pair this with a merit floor — without one, the search can select a genuinely poor overall performer purely because they uniquely cover one target block. Run with enough repeats for a trustworthy stability check and always compare the result against truncation selection's own shortlist, not in isolation.

1.7. Family- or Genetic-Cluster-Quota Selection — best groups, then best lines within them

What it is. Many real programme shortlists do not naturally take the shape of either "top N by value" or "GA-searched coverage set" — they take the shape of "our best few families, and our best few lines out of each." This tool implements exactly that shape directly, rather than checking group balance only as an afterthought once a shortlist already exists.

What question it answers. "Which are my best-performing groups, and what are the best individual lines within each of them?"

Breeding-programme analogy. Instead of admitting the highest-scoring students in the whole school regardless of which classroom they come from, this is closer to admitting the top few students from each of your best classrooms — guaranteeing every strong classroom is represented, rather than letting one especially strong classroom fill every seat.

Two ways to define a "group." A group can be your own pedigree or cross-ID labels (the default, and the natural choice for most programmes), or it can be a genetic cluster built directly from marker-based relatedness data — for programmes where pedigree records are not a fully reliable proxy for actual relatedness (incomplete records, historical mislabelling, or loosely applied family IDs can all decouple "same recorded family" from "actually closely related," and lines with different family labels can still turn out to be closely related, for example through a shared grandparent). When genetic clustering is used, both the pedigree label and the genetic-cluster label are still reported for every selected individual, so you can cross-reference one against the other regardless of which one drove the selection decision.

Why group ranking matters, and the correction applied to it. A small group's raw average score is a noisier, less trustworthy estimate of that group's true quality than a large group's raw average — exactly the kind of small-sample noise classical quantitative genetics corrects for through family evaluation. By default, this tool applies that correction: each group's ranking score is pulled part-way toward the overall average, in proportion to how much that specific group's own size and internal spread justify trusting its raw number at face value. A group with very few members is pulled hard toward the pack; a large group's genuine average is trusted almost fully. This is a meaningful behavioural change from earlier versions of this tool, which ranked purely on each group's raw average regardless of its size — the historical, unshrunk ranking remains available as an option for anyone who specifically wants it.

A second, separate correction: removing the built-in inflation of "best of the bunch." Averaging a group's own best few members has a subtler problem than trust alone, and it is not fixed by shrinkage. Even with perfectly measured scores and zero noise, picking the top performers out of a pool and averaging them will systematically overstate that group's true underlying quality — purely because the more members a group has to choose its "best few" from, the more chances it had to turn up a standout. This is the same reason the tallest student in a class of 200 tends to look taller, relative to that class's real average height, than the tallest in a class of 20, even when every measurement is exact. By default, this tool corrects each group's raw top-members average for exactly this inflation, using the known mathematical expectation of the effect (order-statistics theory), before shrinkage is applied on top of the corrected value — a more

robust treatment of the family mean than shrinkage alone would give. The two corrections pull in different directions and catch different failure modes: shrinkage pulls small, noisy groups back toward the pack; this correction pulls back the flattering, cherry-picked top-average of *any* group that had more room to cherry-pick from — which is sometimes the larger, seemingly more trustworthy ones. Applying both together, as this tool does by default, corrects for more than either one would catch alone.

How many of a group's own top members that ranking is actually based on. A separate setting, independent of shrinkage, decides "what" gets shrunk in the first place: a group's own single best line, or an average of its several best lines. By default, this number matches your quota under the fixed-count option described below — the natural link between "how many lines you are about to take from this group" and "how many of its lines you should judge the group on." The other three ways of deciding how many lines to take from each group, described below, have no equivalent quota to borrow that number from, so by default a group is ranked on its single best member alone under those options. This is worth setting deliberately rather than leaving to the default: switching from the fixed-count option to a percentage or a merit-bar option changes not only how many lines each group contributes, but, unless you say otherwise, the ranking itself.

An optional relatedness ceiling within each group. Just as GA parent selection can be given a relatedness dial (Section 2.2), this tool can too, on the identical 0-90 scale: filling a group's quota then stops being pure score-order truncation and instead balances score against a relatedness ceiling computed from that group's own members — interpolated between that group's naturally best-scoring subset and a proven diversity-maximising reference restricted to the same group. This never discards a group's best remaining member purely to satisfy the relatedness preference; if a slot genuinely cannot be filled without breaching the ceiling, it falls back to score order for that slot rather than leaving it empty.

An optional cross-group diversity check. Picking the single best line from group A and the single best line from group B by score alone says nothing about whether those two lines are genomically complementary — both could, by chance, carry the same favourable region and nothing else new. An optional adjustment addresses this directly, working through the ranked groups in order and preferring, within each group, a line that adds real new coverage relative to everything already claimed by higher-ranked groups' picks — never discarding a group's best line for a marginal diversity gain, but flagging cases where every remaining option in a group was already fully redundant with what came before. Two variants are available: a more rigorous check that looks at total coverage across every important region at once, and a simpler, coarser check that only asks whether a line's single strongest region overlaps with an already claimed one.

Uneven quotas. Quotas per group do not have to be flat — a programme can ask for, say, four lines from its strongest family and only one from a weaker one, rather than the same fixed number from every chosen group.

Four ways to decide how many lines to take from each group. A fixed count (the default, described above) is only one way to answer "how many lines from this group." Three further options are available, each suited to a different kind of programme decision:

- **A fixed count** (the default). "Take the best 3 lines from each chosen group" — the same number regardless of how large or small the group is (subject to uneven quotas per group, above).
- **A percentage of each group's own size.** "Take the best 10% of each chosen group," so a 30-line family contributes 3 lines and a 10-line family contributes 1, automatically scaling with how much genuine within-group selection each group can actually support. A group with only one eligible member is skipped outright under this option — taking "10%" of a single line is not a real selection

decision, it is just taking that one line regardless of merit, so the tool excludes it rather than pretending a quota was applied.

- **A merit bar set relative to the whole population.** Instead of a count or a percentage, take every line — however many that turns out to be — whose score clears a bar set a chosen number of standard deviations above the overall population average. A strong group might contribute five lines this way; a weaker group might contribute none at all.
- **A merit bar set relative to a specific check or reference line.** The same "take everyone who clears the bar" idea, but the bar is defined as a percentage margin above one specific reference line's performance (for example, "at least 10% better than our current check variety") rather than the population average — useful when the real decision-relevant comparison is against a known standard, not an abstract population statistic. This is an inclusive bar: a line beating the check by 20% or 30% still clears a 10% bar just as surely as one beating it by exactly 10%.

What happens when a top-ranked group has nobody clearing its own bar. Under either of the two merit-bar options above, it is possible for one of your top-ranked groups to simply have no member that clears the bar at all — every one of its lines falls short. Rather than counting that group toward your target anyway with zero actual contributors, the tool skips it and moves down to the next-best-ranked group instead, continuing until it has assembled the number of genuinely contributing groups you asked for (or until it genuinely runs out of groups to try, in which case it tells you plainly how many it managed to find instead of silently returning fewer than requested).

When to reach for it.

- Your programme's real shortlist decision already naturally has the shape "our best few groups, and our best few lines out of each" — common in programmes organised around discrete biparental or half-sib families.
- You want that structure enforced directly by the selection rule itself, rather than checking group balance only as a diagnostic after running one of the other tools in this guide.
- You are concerned that pedigree labels alone might not be a fully faithful record of actual relatedness and want a data-derived alternative grouping available alongside — or instead of — pedigree.

When to be cautious or skip it.

- If your programme does not organise around discrete groups in the first place, or a strict per-group quota is not actually something you want enforced, one of the other tools in this guide is a better fit.
- The shrinkage correction is a well-justified, standard statistical approach, but it is still a correction to an estimate, not a guarantee — when every eligible group is very small (for example, exactly one member each), there is not enough information to estimate the correction at all, and the tool falls back to the historical unshrunk ranking with a clear notice rather than silently producing a misleading number.

How it works in practice. You supply your trusted score, your grouping (pedigree labels, or the ingredients for genetic clustering), how many groups to select from, and how many lines to take from each. Everything else — the ranking correction, the relatedness ceiling, the diversity check, uneven quotas — is optional and off by default, so the simplest use of this tool is exactly as straightforward as truncation selection, just organised by group instead of by the whole population at once.

1.8. Usefulness Criterion — ranking crosses, not just parents

What it is. Every tool so far answers "who are the parents." This one answers the next question down: "which specific pairings between shortlisted parents are actually worth making." It ranks candidate crosses — not candidate individuals — by the predicted quality of the best progeny that cross could plausibly produce: roughly, the average value of the two parents, plus a bonus reflecting how much genetic variation that specific cross is predicted to segregate.

What question it answers. "Given my shortlisted parents, which specific pairs should I actually cross?"

Breeding-programme analogy. Two parents with only average individual merit, but who carry different and complementary strengths, can produce better offspring on average than two individually excellent parents who happen to be very similar to each other. This tool is built specifically to catch that — it is not simply "cross your two best-ranked parents together and stop there."

Why a high-ranking cross does not need two high-ranking parents. A cross's predicted segregation variance term rewards genetic complementarity, not just average merit. A cross between a strong parent and a complementary — but individually more moderate — partner can out-rank a cross between two similar, individually stronger parents, because it is predicted to produce a wider and better-tailed spread of offspring to select from afterward.

Multiple ways to estimate a cross's potential. Several methods are available, trading off how much data you need against how rigorous the estimate is — from a fast method that reuses the same block-level breakdown already produced earlier in the pipeline, through methods that use phased haplotype data for more exact within-block detail, up to a fully rigorous multi-locus method suited to inbred-line (doubled-haploid or recombinant inbred line) breeding programmes specifically. Match the method to the data you already have rather than reaching for the most demanding one by default — the fastest method is the right starting point for most panels, and it is worth checking whether a more rigorous method actually changes your top-ranked crosses before investing in the extra data requirements.

When to reach for it.

- You already have a shortlist of candidate parents (from any of the tools above, or built any other way) and need the next-level decision: not "who do I keep," but "which pairs do I actually cross."
- Your programme only has capacity to make a limited number of biparental crosses per cycle and wants to prioritise the ones most likely to segregate a genuinely better best-progeny tail, not simply the crosses between the two highest-ranked individuals.
- You suspect a complementary-but-moderate pairing might outperform two similar elite parents as a cross — for example, where the two parents carry different favourable regions rather than the same one.

When to skip it.

- If you only need a parent list, not a cross list, this adds nothing — a tool from Sections 2.1-2.3 is enough on its own.
- If you need actual contribution numbers and a full mating plan balanced across your whole set of crosses at once, with population-wide inbreeding under an explicit cap, this is not that tool — it ranks each candidate cross independently, with no notion of how the crosses relate to each other as a whole plan. Section 2.5 is built for that.

How it works in practice. Feed it your shortlisted parent list and their breeding values, choose how large a fraction of each cross's progeny you would realistically keep (this drives how much weight is given to a cross's predicted spread, not just its average), and choose which variance-estimation method matches the data you have. The output is a ranked table of candidate crosses — worth inspecting directly to see whether the top-ranked cross is simply the two highest-merit parents, or something more genuinely complementary.

1.9. Optimal Contribution Selection — a full mating plan, with inbreeding under control

What it is. This is the tool for turning a shortlist into an actual mating plan: how much each parent should contribute, and which specific matings should happen, to maximise expected genetic merit subject to an explicit, negotiated cap on the resulting population's relatedness. This is the classical Optimal Contribution Selection (OCS) problem from quantitative genetics, solved here via a choice of three mature, external solving engines rather than a method invented specifically for this package.

What question it answers. "How much should each parent contribute, and who should mate with whom, so that I get the best possible genetic progress without letting relatedness in the resulting population creep past a line I have chosen not to cross?"

Breeding-programme analogy. Rather than deciding up front "each of my 20 shortlisted parents gets exactly one cross," this tool decides that your strongest, most complementary parents should contribute more crosses than your merely-adequate ones — but caps how far that concentration can go, because contributing too heavily also means contributing too much relatedness into the next generation.

Three engines, one shared dial. All three available engines share a single relatedness-vs-gain dial on the same 0-90 scale used throughout this guide (0 prioritises maximising merit, accepting more relatedness; 90 prioritises minimising relatedness) — but they reach that goal through genuinely different mechanisms and have different practical requirements. Two of the three genuinely solve the full contribution-and-mating optimisation problem; the third instead performs a related but distinct task — a fast, discrete, greedy method for predicting and selecting good crosses, which is a different algorithm class, not a lesser version of the same one. Which engine is right depends mostly on what software your programme already has installed and whether you specifically want that third engine's own discrete cross-selection approach rather than a true contribution optimum.

When to reach for it.

- You are running an ongoing population-improvement or recurrent-selection programme, where inbreeding accumulation across cycles is a real, actively managed risk rather than a one-off concern.
- You need an actual mating list with contribution numbers, not just a shortlist of parents or a ranked list of individual crosses.
- Your programme maintains several breeding pools long-term, where "how much should each parent contribute" is a question with consequences well beyond the current cycle alone.

When to skip it.

- For a handful of one-off biparental crosses in an early-generation line-development programme, the cross-ranking tool (Section 2.4) is usually enough, and a formal, population-wide inbreeding cap is not yet the binding constraint.
- If none of the three engines' required software is available and installing one is not practical right now, the relatedness dial built directly into GA parent selection (Section 2.2) is a softer, already available way to manage relatedness at the shortlist stage, without needing any additional software.

How it works in practice. Supply your trusted merit values and a relationship matrix for your candidates and choose a starting point on the 0-90 dial — 30 is a reasonable, moderate first pass for most programmes. If the resulting plan's relatedness looks too high for your programme's tolerance, move the dial toward 90; if it looks too conservative and you suspect merit is being left on the table, move it toward 0. All engines respect a cap on how many crosses any single parent can contribute to, and guarantee that the same specific mating is never repeated in the plan.

Does this tool select its own parents, or only work with the shortlist you hand it? Both, in a sense worth being precise about. This tool can never choose a parent who wasn't in the candidate list you supplied — in that respect it is strictly downstream of whichever shortlist tool you used. But it is not required to give every supplied candidate a role in the final plan, and you can ask it to actively narrow further, via an optional cap on the total number of distinct parents used. The three engines handle that cap very differently, and the difference matters if you rely on it:

- **AlphaMate** enforces the cap natively, as an integral part of its own search — it decides simultaneously which subset of your candidates contributes at all, how much each contributes, and who mates with whom, all in one optimisation. If you hand it 20 candidates and cap the plan at 12 parents, AlphaMate itself chooses which 12 — this is a genuine, built-in selection step happening *inside* the mating-plan tool, not merely an allocation exercise over a fixed roster you already finalised.
- **The optiSel-based engine** instead solves the contribution problem over your *entire* supplied list first, with no cap, and only afterward — as a separate, non-re-optimised patch — zeroes out every contribution below the cutoff needed to meet your requested cap. This gets to a similar place but by a cruder, two-step route rather than one joint search.

- **The SimpleMating-based engine** does not support this cap at all; it is simply ignored. Its own cross-selection process can still leave some supplied candidates without a role in the final plan, just not because of a deliberate count-based cap.

The practical takeaway: your shortlist is always a hard ceiling on who *can* appear in the final mating plan, but it is not a guarantee that everyone on it *will* — and if you specifically want the mating-plan tool itself to make that further narrowing decision well, the AlphaMate engine is the one built to do that job properly, as one integrated optimisation rather than a shortlist step followed by a separate allocation step.

1.10. Pareto Frontier Sweep — seeing the trade-off, not guessing a single number

What it is. The relatedness dials described in Sections 2.2 and 2.5 each collapse the merit-vs-relatedness trade-off into a single number you have to choose. This tool instead runs the underlying search repeatedly across a whole range of settings and keeps only the genuinely best trade-off points — the settings where no other setting tried achieved both higher merit *and* lower relatedness at the same time — so you can see the actual empirical trade-off curve and choose a point on it deliberately, rather than committing to one guessed number up front.

What question it answers. "What does my programme's actual gain-versus-relatedness trade-off look like, across the whole range of possible choices — not just at the one setting I happened to try?"

Breeding-programme analogy. Rather than picking one point on a map and hoping it is a good one, this draws the whole coastline first, so you can see exactly what you would be giving up in merit for any given amount of relatedness reduction, and choose your point on that coastline with full information.

When to reach for it.

- You are not confident what a specific dial setting should be for your programme, and want to see the actual trade-off shape before committing to a single number — especially useful the first time you are setting this policy, or when you need to justify a specific choice to a review committee or collaborator (an empirical trade-off curve is a stronger argument than "we used this number because it seemed reasonable").
- Periodically, even in an established programme, to check whether a previously chosen setting still sits in a sensible place on the current cycle's trade-off curve — the shape of that curve can shift as your candidate pool changes over time.

When to skip it.

- If your programme already has institutional experience with a specific setting that has worked well before, re-running this sweep every single cycle has a real computational cost for limited added information — this is not intended as a routine per-cycle step.
- Under time pressure, a single run of the underlying search with a sensible default setting is often good enough.

How it works in practice. You provide the same inputs the underlying search would need, plus a range of relatedness-weight settings to try across. The output is a table showing, for each setting actually worth considering, the realised merit and realised relatedness of the resulting founder set — plot or scan this table by eye, pick the point that matches your programme's actual risk tolerance, and pull that specific run's parent list out for use.

1.11. Exact Validation — checking a heuristic mating plan against the true optimum

What it is. Every mating-plan tool in this guide, including the search methods underlying Sections 2.2 and 2.5, is a heuristic in a specific sense: none of them is mathematically guaranteed to find the single best possible discrete mating plan (a continuous contribution-level optimum can be solved exactly by one of Section 2.5's engines, but converting that into an actual discrete list of specific matings, while respecting per-parent and no-repeat constraints, is still solved approximately). This tool solves the same underlying cross-selection problem exactly, via a different optimisation technique (integer programming) suited specifically to finding guaranteed-optimal solutions on small-enough problems — so a heuristic plan's quality can be checked against a genuine, provable optimum, not just trusted on faith.

What question it answers. "How good is my heuristic mating plan, really — how far is it from the best plan that was actually achievable given my constraints?"

Breeding-programme analogy. This is a spot-check, not a production method — closer to independently re-solving a scheduling problem by brute force on a small test case to confirm the scheduling software's answer was actually close to optimal, rather than trusting the software's output blindly on every real, much larger case going forward.

When to reach for it.

- You want a documented, defensible answer to "how good is our heuristic mating plan, really" — for internal quality assurance, a funder or reviewer report, or before formally adopting a new mating-plan tool as your programme's standard.
- Your candidate cross list, after any reasonable pre-filtering, is small enough to solve exactly — this method does not scale to the size of a full, unfiltered candidate list.
- You are deciding between two different heuristic plans and want a common, independent yardstick to compare them against, rather than comparing their own self-reported scores against each other.

When to skip it.

- This is not a routine, cycle-to-cycle production step. For day-to-day mating-plan generation, use Section 2.5's engines directly.
- Do not reach for it on a large, unfiltered candidate set — pre-filter first (a tighter relatedness cutoff, or restricting to your top-ranked crosses) rather than forcing an oversized problem through a method that is not built to scale that far.

How it works in practice. Feed it the same cross-ranking table you would otherwise hand to a heuristic tool, matching whatever constraints your real plan used (how many crosses total, how many any one parent can appear in), and compare the exact solution's score against your heuristic plan's own score. A large gap between the two is a signal to reconsider the heuristic tool or its settings; a gap near zero is itself a useful, reassuring result — it means the heuristic left little to no achievable gain on the table for that particular candidate set.

1.12. Core Collection — building a diverse founder set, when diversity itself is the goal

What it is. Every other tool in this guide treats diversity as a constraint layered on top of a merit objective. Sometimes the actual goal is the reverse: a genebank core collection, a broad training or reference panel, or a deliberately diversification-focused founder set, where representing the full breadth of genetic diversity in a panel **is** the objective, not a side constraint to be managed around a merit ranking. This tool selects a subset that spreads as widely as possible across the available genetic diversity, with an optional minimum-merit floor so you can still ask for "the most diverse subset among candidates that already clear an acceptable bar."

What question it answers. "Which subset of this panel best represents its overall genetic diversity?"

Breeding-programme analogy. Rather than admitting the strongest candidates, this is closer to deliberately assembling a representative cross-section of the whole population — useful when the goal is representing breadth itself, such as building a reference collection for future use, not ranking today's best performers.

When to reach for it.

- Genebank or germplasm curation — choosing a representative core subset from a large accession collection for long-term conservation or characterisation.
- Building or refreshing a broad training or reference population for genomic prediction, where representing the panel's full diversity matters more than merit.
- Starting a new diversification or pre-breeding programme from a wide genetic base, deliberately before any selection pressure is applied.

When to skip it.

- If you are choosing crossing parents for the next cycle of an active improvement programme, merit matters there and this tool optimises for diversity alone — the merit floor here only sets a minimum bar, it does not rank by merit above that bar the way every other tool in this guide does. Sections 2.2, 2.4, and 2.5 are the merit-driven tools for that job.
- Worth pausing on if your relationship matrix is not yet trustworthy (poor marker density, questionable ploidy assumptions) — this tool's selection is only as good as the distances it is given to work with.

How it works in practice. Choose whether avoiding near-duplicate accessions is the priority (the default approach) or whether you want the selection spread evenly across the whole diversity space rather than anchored on the most extreme outliers (an alternative approach). The output includes a step-by-step trace of how diversity accumulated as individuals were added — useful for judging whether your chosen collection size is already large enough, or whether the marginal diversity gain per added individual is still climbing and a larger collection would be worthwhile.

Part 3 — Putting it together

1.13. Decision guide

If your question is...	Reach for...
"What's the simplest defensible parent shortlist?"	Truncation Selection
"Are my best-by-value individuals actually covering my target genetic regions, or duplicating the same ones?"	GA Parent Selection
"I want block coverage, but I'm worried the search might be selecting on noisy estimates."	GA Parent Selection's merit dial
"I want a relatedness cap on my founder set, without guessing a raw penalty number."	GA Parent Selection's relatedness dial (or Optimal Contribution Selection's, for a full mating plan)
"My programme's shortlist is naturally 'best few families, best few lines each.'"	Family- or Genetic-Cluster-Quota Selection
"I'm not sure my pedigree records are a reliable proxy for actual relatedness."	Family- or Genetic-Cluster-Quota Selection, genetic-cluster mode
"Which specific pairings among my shortlisted parents are worth making?"	Usefulness Criterion
"How much should each parent contribute, and who mates whom, with inbreeding under an explicit cap?"	Optimal Contribution Selection
"What does the actual gain-vs-relatedness trade-off curve look like, rather than one setting?"	Pareto Frontier Sweep
"How close is my heuristic mating plan to the true best achievable plan?"	Exact Validation
"I want a diverse founder or reference panel, not a merit-ranked shortlist."	Core Collection

1.14. A rough guide by programme shape

Population-improvement or diversity-management programmes — many families, ongoing recurrent selection across overlapping generations — tend to lean most on Optimal Contribution Selection and the Pareto Frontier Sweep, to manage population-wide inbreeding deliberately across cycles rather than cycle by cycle in isolation.

Programmes built around a small number of elite biparental families, pushed toward rapid fixation, tend to lean more on GA Parent Selection and the Usefulness Criterion, to pick a tight, genuinely complementary parent and cross set rather than managing a large, ongoing population.

Programmes organised around discrete families or crossing groups, where "best few groups, best few lines each" is already the natural shape of the decision, should start with Family- or Genetic-Cluster-Quota Selection directly, rather than checking group balance only after the fact.

Core Collection selection serves a different need again — building or refreshing a diverse base population — and Exact Validation is a check you can run against any of the other mating-plan tools' output, not a competing strategy of its own.

1.15. A word on running more than one tool

None of the tools in this guide is a strict improvement on the others, and several answer genuinely different questions from each other. In practice, the strongest evidence for a crossing decision usually comes from running more than one and comparing: how much overlap is there between a plain ranked shortlist and a block-aware one? Does a family- or cluster-balanced shortlist look meaningfully different from an unconstrained one? Where two methods agree strongly, that agreement is itself reassuring; where they disagree, that disagreement is worth investigating directly rather than picking whichever answer happened to come first.

Glossary

Breeding value / GEBV. A predicted measure of an individual's genetic merit for a trait, typically estimated from a genomic prediction model. "Local GEBV" refers to the portion of that value attributable to a specific haplotype block, rather than the whole genome at once.

Coancestry / relatedness. How closely related two individuals, or the members of a chosen group, are to each other. Managing relatedness in a crossing decision is about limiting how much inbreeding risk accumulates in the resulting offspring or future population.

Genetic cluster. A group of individuals identified as closely related directly from marker-based relatedness data, rather than from recorded pedigree/family labels — used as an alternative or complementary grouping in family-quota selection.

Haplotype block. A short, co-inherited run of the genome, treated as one unit for the purposes of tracking which favourable genetic material a candidate carries and where.

Optimal Contribution Selection (OCS). The classical quantitative- genetics problem of deciding how much each candidate parent should contribute to the next generation, and which specific matings should happen, to maximise expected genetic gain subject to an explicit cap on population-wide relatedness.

Pareto-optimal / trade-off frontier. A point in a trade-off (such as merit versus relatedness) is Pareto-optimal if no other option achieves better performance on *every* dimension at once. A trade-off frontier is the full set of such points — the genuine "best available" curve, rather than one guessed setting.

Relationship matrix (GRM). A matrix recording how closely related every pair of candidates is, typically built from marker data (or, less commonly, pedigree records). Higher values mean closer relatedness; the matrix's diagonal reflects an individual's own inbreeding level.

Shrinkage / small-sample correction. A statistical correction that pulls a small group's estimated value part-way toward the overall average, in proportion to how little that group's own data can be trusted on its own — used by default in family- and genetic-cluster-quota selection's group ranking.

The 0-90 scale. A shared convention used across several tools in this guide for expressing a merit-vs-relatedness preference: 0 means prioritise merit/gain and accept more relatedness; 90 means prioritise minimising relatedness, even at some cost to merit. The same scale means the same thing everywhere it appears in this guide, even though different tools convert that preference into an actual constraint through different underlying mechanisms.

Where to go next

This guide is deliberately non-technical and code-free. For the full statistical detail behind every tool described here — equations, worked numerical examples, and the exact arguments and options each function accepts — see the companion vignette, *From Local GEBV to a Crossing Decision*, and each function's own detailed help page within the package.